Use Case: data-driven camera calibration for light field camera systems

Author: Youssef Daoudi El Boukhrissi (4277022)

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Use Case: Cross-Shaped Camera Array [Center + North/East/South/West]

Phase 1 — Initial Reference Calibration (once)

- **Target-based, OpenCV:** Perform a one-time checkerboard-based calibration to estimate intrinsic and extrinsic parameters for all cameras of the cross-shaped array.
- **Optional refinement:** Apply an optional global refinement (bundle adjustment) to improve consistency of the estimated parameters.
- **Baseline creation:** Compute a global health score for the initial calibration and store the result as `baseline.json`. This baseline represents the known-good reference calibration for later recalibration decisions.
- **Logging:** Store the full initial calibration as a timestamped JSON file and write a summary line to `calibration.log`.

Phase 2 — Automated Recalibration Attempts (periodic scheduler)

- **Trigger:** A fixed time interval (e.g. every 5 minutes) executes an automated recalibration cycle.
- **Conditional data preparation (optional):** If LiFCal-compatible data (`LiFCal_Data` with focus and depth directories) is already present and complete, the system reuses it. Otherwise, the system runs `run_imageset_creation.py` to generate the required inputs (focus images and virtual depth).
- **LiFCal recalibration run:** Execute `run_LiFCal_calibration.py` which:
  - runs LiFCal recalibration for each camera (Center, Up/Down/Left/Right),
  - exports per-camera parameter JSON files (`parameters.json`),
  - creates a combined multi-camera result (`combined.json`).
- **Logging and traceability:** Store the combined LiFCal result as a timestamped JSON file in `/data/baseline` and append all relevant run information to `calibration.log`.

Phase 3 — Health Evaluation and Baseline Update Rule (commit decision)

- **Health score (LiFCal):** Compute a single global health score (range $[0, 100]$) from the LiFCal combined parameter result (`combined.json`). The score represents the calibration quality of the complete camera array for this recalibration cycle.
- **Comparison against baseline:** Compare the new LiFCal health score with the health score stored in the baseline calibration.
- **Acceptance rule (commit):** If the new health score is strictly higher than the baseline score, replace `baseline.json` with the new combined calibration result. Otherwise, keep the existing baseline unchanged.

- **Continuous improvement loop:** The scheduler repeats the recalibration cycle periodically. Over time, improved recalibration runs can update the baseline, enabling a data-driven refinement process without manual intervention.